

Data Structure (C Programming)

03/09/2026

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Roll No : 15

Batch : B1

Assignment No 3

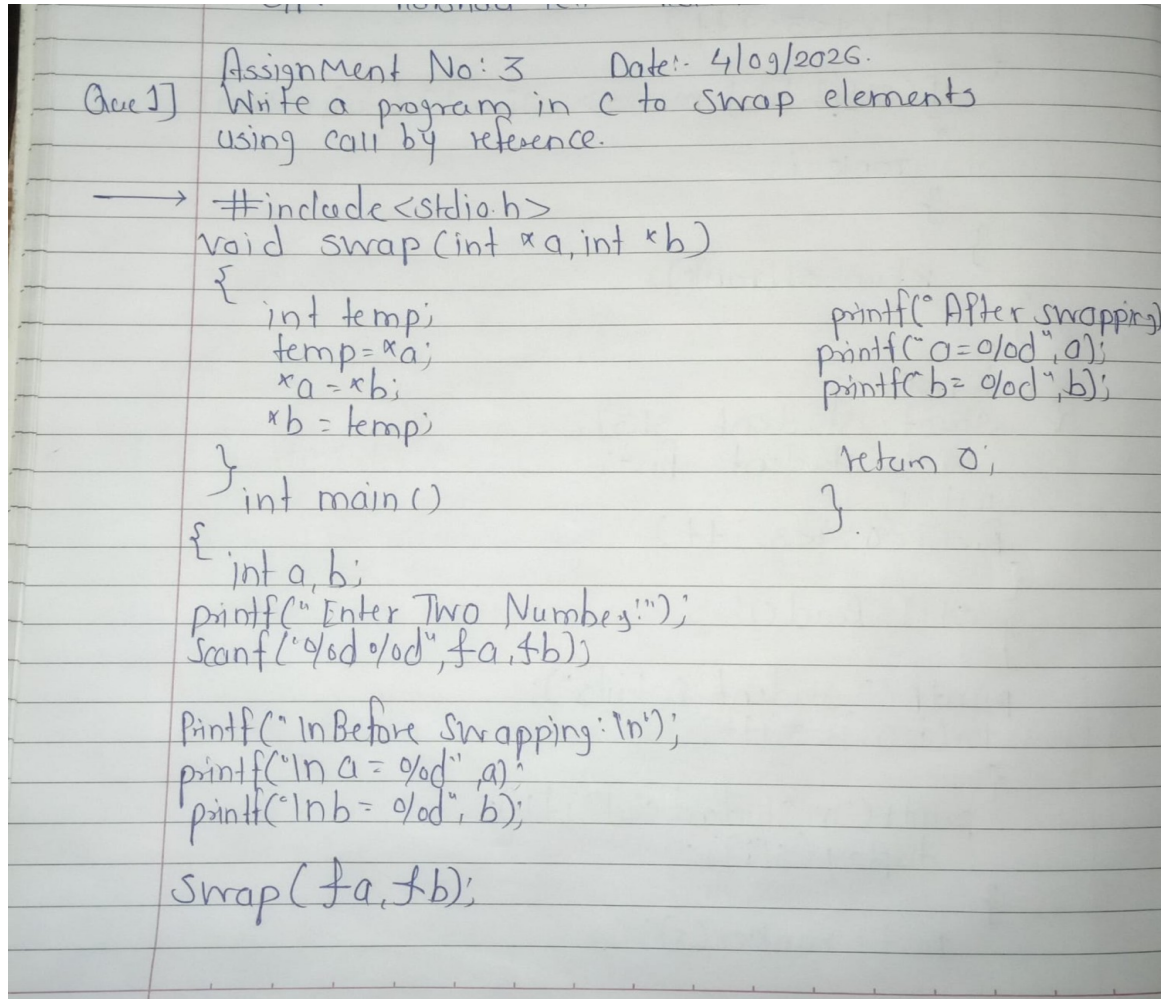
Pointers.

Write a C program for following problem statements:

1. Write a program in C to swap elements using call by reference.
2. Write a program in C to compute the sum of all elements in an array using pointers.
3. Create a structure called "Student" with members name, age, and total marks. Write a C program to input data for two students, display their information, and find the average of total marks. **Use a Structure pointer.**

Q1)

Input:



Assignment No: 3 Date: 4/09/2026.

Ques 1] Write a program in C to swap elements using call by reference.

```
#include <stdio.h>
void swap(int *a, int *b)
{
    int temp;
    temp = *a;
    *a = *b;
    *b = temp;
}

int main()
{
    int a, b;
    printf("Enter Two Numbers:");
    scanf("%d %d", &a, &b);

    printf("In Before Swapping: \n");
    printf("In a = %d", a);
    printf("In b = %d", b);

    swap(&a, &b);

    printf("After swapping: \n");
    printf("a = %d", a);
    printf("b = %d", b);

    return 0;
}
```

Output:

```
student@student-OptiPlex-SFF-7020:~/Videos/Assignment_No3$ gcc -o output Call_by_references.c
student@student-OptiPlex-SFF-7020:~/Videos/Assignment_No3$ ./output
Enter first number: 24
Enter second number: 30

Before swapping:
a = 24
b = 30

After swapping:
a = 30
b = 24student@student-OptiPlex-SFF-7020:~/Videos/Assignment_No3$
```

2] Write a program in C to compute the sum of all elements in an array using pointers.

```
#include <stdio.h>
int main()
{
    int arr[100], n, i;
    int sum = 0;
    int *ptr;
    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements: \n", n);
    for(i=0; i<n; i++)
    {
        scanf("%d", &arr[i]);
    }
    ptr = arr;
    for(i=0; i<n; i++)
    {
        sum = sum + *(ptr+i);
    }
    printf("sum of all elements = %d", sum);
    return 0;
}
```

O/P: Enter number of elements: 5
Enter 5 elements
2 4 6 8 10
Sum of all elements = 30.

Output:

```
student@student-OptiPlex-SFF-7020:~/Videos/Assignment_No3$ gcc -o output sum_of_elements.c
student@student-OptiPlex-SFF-7020:~/Videos/Assignment_No3$ ./output
Enter 5 elements:
3
6
2
8
9
Sum of all elements = 28student@student-OptiPlex-SFF-7020:~/Videos/Assignment_No3$
```

- 3] Create a Structure called "Student" with members name, age, and total marks. Write a C program to input data for two Students, display their information, and find the average of total marks. Use a Structure pointer.

```
→ #include <stdio.h>
struct Student {
    char name[50];
    int age;
    float total_marks;
};

int main()
{
    struct Student S[2];
    struct Student *ptr;
    int i;
    float sum=0, average;
    ptr=S;

    for(i=0; i<2; i++) {
        printf("Enter details of student %d:\n", i+1);
        printf("Enter Name:");
        scanf("%s", (ptr+i)->name);

        printf("Enter Age:");
        scanf("%d", &(ptr+i)->age);

        printf("Enter Total Marks:");
        scanf("%f", &(ptr+i)->total_marks);
    }
```

Display information.

```
for(i=0; i<2; i++)
```

```
{
```

```
    printf("Student o/d: \n", i+1);
```

```
    printf("Name: %s\n", (ptr+i)->name);
```

```
    printf("Age: %d\n", (ptr+i)->age);
```

```
    printf("Total Marks: %.2f\n", (ptr+i)->totalmarks);
```

```
    sum = sum + (ptr+i)->totalmarks;
```

```
}
```

```
average = sum/2;
```

```
printf("Average of Total Marks= %.2f\n", average);
```

```
return 0;
```

```
}
```

O/p:

Enter details of Student 1:

Enter Name: Rahul

Enter Age: 20

Enter Total Marks: 450

Enter details of Student 2:

Enter Name: Harshad

Enter Age: 19

Enter Total Marks: 480.

-- Student Info --

Student 1:

Name: Rahul

Age: 20

Total Marks: 450.00.

Student 2:

Name: Priya

Age: 19.

Total Marks: 480.00.

Average of Total

Marks = 465.00.

Harshad Teli

Roll No: 15



Output:

```
student@student-OptiPlex-SFF-7020:~/Videos/Assignment_No3$ gcc -o output structure3.c
student@student-OptiPlex-SFF-7020:~/Videos/Assignment_No3$ ./output

Enter details of Student 1
Enter name: Harshad
Enter age: 21
Enter total marks: 450

Enter details of Student 2
Enter name: Shrinath
Enter age: 19
Enter total marks: 560

----- STUDENT DETAILS -----

Student 1
Name: Harshad
Age: 21
Total Marks: 450.00

Student 2
Name: Shrinath
Age: 19
Total Marks: 560.00

Average of total marks = 505.00student@student-OptiPlex-SFF-7020:~/Videos/Assignment_No3$
```

